Ian Abiel Wangsa

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Education

National University of Singapore (NUS) — Singapore

B.Sc. in Computer Science (Minor in Economics & Statistics) | ***GPA: 4.50/5.00*** — *Aug 2025 – Aug 2029*

- **Coursework:** Programming Methodology I & II, Data Structures & Algorithms, Linear Algebra, Discrete Structures, Calculus.

Eunoia Junior College — Singapore

GCE A-Levels (Science Stream: H2 Math, Physics, Chemistry, Economics) — *Jan 2021 – Nov 2022*

- **Academic Excellence:** Awarded Dean's List (2021, 2022). Received Edusave Merit and Academic Achievement Awards.
- **Varsity Softball:** 2nd Place in National School Games (2022); Awarded National Colours Award.

Experience

Caerus Global Management — Singapore

Quantitative Summer Analyst – Apprenticeship — *Aug 2026 – Present*

- Architected an asynchronous streaming pipeline using `websockets`, `aiohttp`, and `orjson` to reconstruct stateful order books with sequence gap validation across Binance USDS-M and Hyperliquid for 100ms cross-venue grid analysis.
- Engineered microprice, Order Flow Imbalance (OFI), and cross-venue basis features; quantified a 0.7–0.8s Binance lead-lag signal over Hyperliquid via lead/lag estimation and walk-forward cross-validation.
- Developed an event-driven backtesting engine modeling ladder-walking market impact, maker/taker fee structures, and execution delays to analyze signal decay and fee friction limits on HFT arbitrage.

Stealth Startup — Singapore

Product Engineer Intern — *Jul 2026 – Present*

- Architected automated E2E regression pipelines with browser-driven test harnesses and artifact-backed QA reports, collapsing manual test cycles across distributed multi-service product surfaces.
- Instrumented product analytics dashboards with SQL-backed KPI queries, cohort segmentation, and scheduled exports to deliver WAU/MAU telemetry and registration funnel snapshots.
- Owned full-stack defect lifecycle from triage to post-merge verification; shipped web features across core user flows while building out React Native mobile app with native sign-in.

CTES Consulting — Singapore

Software Engineer - Off-Cycle Intern — *Jul 2026 – Present*

- Engineered full-stack profiling and data extraction pipelines using React, TypeScript, Tailwind CSS, Node.js, and PostgreSQL, deployed on AWS behind Cloudflare DNS/Security layers.
- Developed automated web scraping and session management modules using Python and Playwright to navigate complex portals and handle FedAuth cookie authentication.
- Built document parsing and report generation engines using `python-docx`, `pypdf`, and `openpyxl`, reducing candidate profile processing turnaround time from **1–2 days to a few hours**.

Accenture — Singapore

Summer Intern – Application Engineer, Technology (Health & Public Services) — *Apr 2026 – Jul 2026*

- Engineered an enterprise Agentic AI Root Cause Analysis Generator using hybrid retrieval pipelines (BGE, BM25) and Azure DevOps integrations to analyze codebases and log traces directly.
- Architected an automated Test Script Generator utilizing Claude CLI Agentic workflows and custom sub-agents, cutting incident testing turnaround time from days to **under 30 minutes** (**95%+ efficiency boost**).
- Built a Flask web frontend powered by a custom MCP Server and Playwright scrapers for SharePoint authentication, while remediating critical production bugs to ensure SLA compliance.

D+A Strategies — Remote

Quant Trainee - Apprenticeship — *Mar 2026 – Present*

- Designed, backtested, and stress-tested systematic algorithmic trading strategies using vectorized mathematical pipelines over unstructured financial time-series data.
- Constructed quantitative modeling environments in Python to evaluate alternative pricing patterns and optimize statistical arbitrage performance.

Republic of Singapore Air Force (RSAF) & Singapore Army — Singapore

Air Force Signals Commander & Specialist Cadet (NSF) — *Mar 2023 – Jan 2025*

- **Key Deployments:** Selected for Exercise Wallaby (Rockhampton, Australia) to establish joint SAF-RAAF radio communications; participated in SAF Day Parade (Jul 2024).
- **Unit Leadership:** Served as Comms Store IC, Gym Innovation IC, and National Education IC, leading 160 SQN to win the NE CDF Award for two consecutive years (2023–2025).

Projects & Hackathons

BrainHack TIL-AI 2026 — DSTA Singapore | *1st Place (Team Pandamonium)* May 2026 – Jun 2026

- **Tech Stack:** Python, Docker, FastAPI, PyTorch, NeMo, YOLOv11, BM25, Transformers, SB3
- Built a finals-ready multimodal AI system with 5 Dockerised FastAPI microservices deployed in an air-gapped environment.
- **ASR:** Fine-tuned NVIDIA Parakeet TDT with custom decoder, FP16 inference, sorted batching, and phrase correction.
- **CV:** Engineered a YOLOv11 target detection pipeline for aircraft, vehicles, and vessels with threshold tuning.
- **Noise:** Built a YOLO-surrogate PGD adversarial noise generator constrained by SSIM/RMSE budgets to fool opponent detectors.
- **NLP:** Deployed a BM25-first RAG pipeline with alternative neural reranking layers and quantized LLM generation.
- **AE:** Developed a MaskablePPO RL navigation agent featuring action masking, reward shaping, and fallback BFS/A* pathing.

AlphaEngine | *Python, Asyncio, WebSockets, Telegram Bot API, Supabase, PostgreSQL, PyArrow, Docker* Jul 2026

- Architected a real-time quant execution system using `websockets`, `aiohttp`, and `orjson` to stream high-frequency L2 order book deltas, depth snapshots, and trade ticks across Binance, Bybit and Hyperliquid.
- Implemented stateful local order book reconstruction with sequence gap validation ($U/u/pu$), automated WebSocket resynchronization, and compressed Parquet storage for 100ms cross-venue grid analysis.
- Engineered a Telegram Bot execution and monitoring add-on, broadcasting automated risk alerts, trade execution updates, and real-time PnL/position telemetry directly to remote operators.
- Integrated a persistent Dockerized gateway server with Supabase PostgreSQL and Row Level Security (RLS) for transactional order blotter logging and strict risk limit management.

Delta1 Trading Strategy | *Python, Pandas, NumPy, SciPy, Statsmodels, Jupyter* July 2026

- Engineered high-frequency microstructure signals including Microprice, Order Flow Imbalance (OFI), and cross-venue basis z-scores; quantified a 0.7–0.8s Binance predictive lead over Hyperliquid.
- Constructed an event-driven backtesting engine modeling full ladder-walking order book market impact, tier-based maker/taker fees, and execution latency sweeps (100ms–1000ms).
- Validated strategy alpha via walk-forward Information Coefficient (IC) analysis, shuffled-signal control benchmarks, and round-trip cost floor decomposition to evaluate HFT arbitrage viability.

Leadership & Co-Curricular Activities

NUS MedTech | *Software Engineer* Aug 2026 – Present

- Developing healthcare and medical technology software solutions, engineering robust frontend and backend integrations.

NUS Student Investment Fund (NUSSIF) | *Quantitative Developer Analyst* Aug 2026 – Present

- Engineering quantitative tooling and automated financial analysis pipelines to support data-driven portfolio management.

Google Developer Group @ NUS | *Software Engineer* Aug 2026 – Present

- Built full-stack platforms and infrastructure for non-profit Caring For Life using modern web tools.

NUS Fintech Society | *Machine Learning Trainee* Aug 2026 – Present

- Researching and building ML models for finance, mathematical predictions, streamlining AI and agentic AI systems.

NUS Games Development Group | *Software Engineer (Pandamonium)* Jan 2026 – Present

- Engineered procedural board rendering and centralized state management in Unity/C# for a 2D game.

NUSSU VENtures | *Operations Executive* Jan 2026 – Present

- Oversee event planning, venue coordination, and cross-functional communication among stakeholders.

NUS Sports Club - RunNUS | *Admin & Finance Executive* Jan 2026 – Sep 2026

- Managed budgeting, financial tracking, and vendor procurement for a run engaging 1,000+ runners.

NUS Sports Club - SunNUS | *Team Sports Series Executive* Jan 2026 – Jun 2026

- Coordinated event logistics, venue setup, and tournament fixtures for a 400+ person beach sports event.

NUS Community Service Club - Project BHID | *Programmes Head* Aug 2025 – Aug 2026

- Spearheaded programmes for the elderly with intellectual disabilities while directing student volunteer training sessions.

Youth Corps Singapore & NUS Rotaract Club | *Volunteer Educator* Aug 2025 – Jan 2026

- Taught Scratch, computational thinking, and foundational AI concepts to primary school students through workshops.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C#, HTML, CSS
Frameworks: React, React Native, Next.js, FastAPI, Node.js, Spring Boot, Flask, Unity, Tailwind CSS, PyTorch, Pandas
Developer Tools: Docker, AWS, Cloudflare, Git, Linux, Azure DevOps, Power Automate, Playwright, MediaPipe, ChromaDB
Certifications: Databricks AI/BI, AWS Cloud Practitioner (CLF-C02), Cisco Specialist, Anthropic Claude 101, GitHub